

DATA FLOW

From Raw CSV to Intelligent Timetable

Course & Room Dataset



Raw Inputs

Preprocessed .csv files for rooms and courses

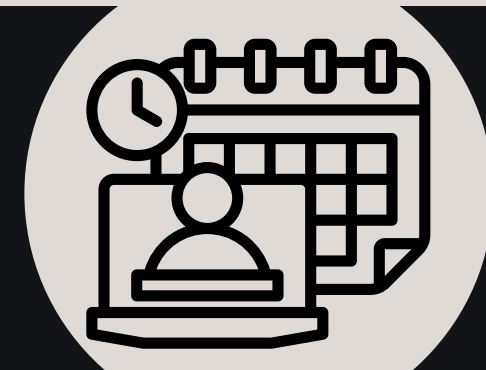
Session Generator



Session Breakdown

Splits courses into lab/theory sessions with duration & section

Time Slot Engine



Slot Matrix Builder

Generates available 90-minute slots for Mon–Fri

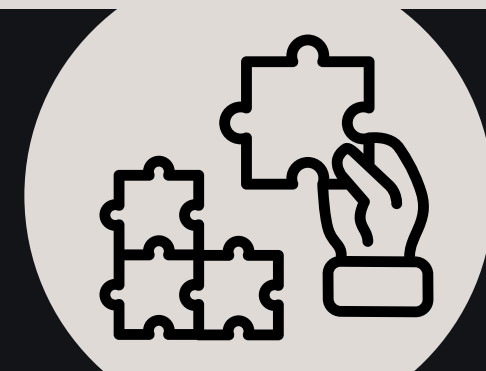
Genetic Algorithm Core



AI Optimizer

Evolves population to schedule sessions with minimal conflict

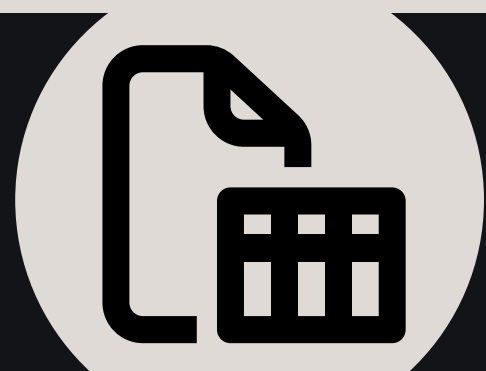
Conflict & Fallback Handling



Smart Placement Logic

Applies forced strategies and logs missed sessions

Timetable Exporter



Output Generator

Produces structured Excel sheets per room, teacher & section

Visual Formatter



Color-Coded Visuals

Generates clear, printable Excel timetables with watermark

Khadijah Zahoor
F2023-956

ClassSync AI

Because Your Time Deserves a Better Schedule.



PROBLEM STATEMENT

Manual university timetable creation is time-consuming, prone to conflicts, and unable to scale with growing class and resource complexity.

PROPOSED SOLUTION

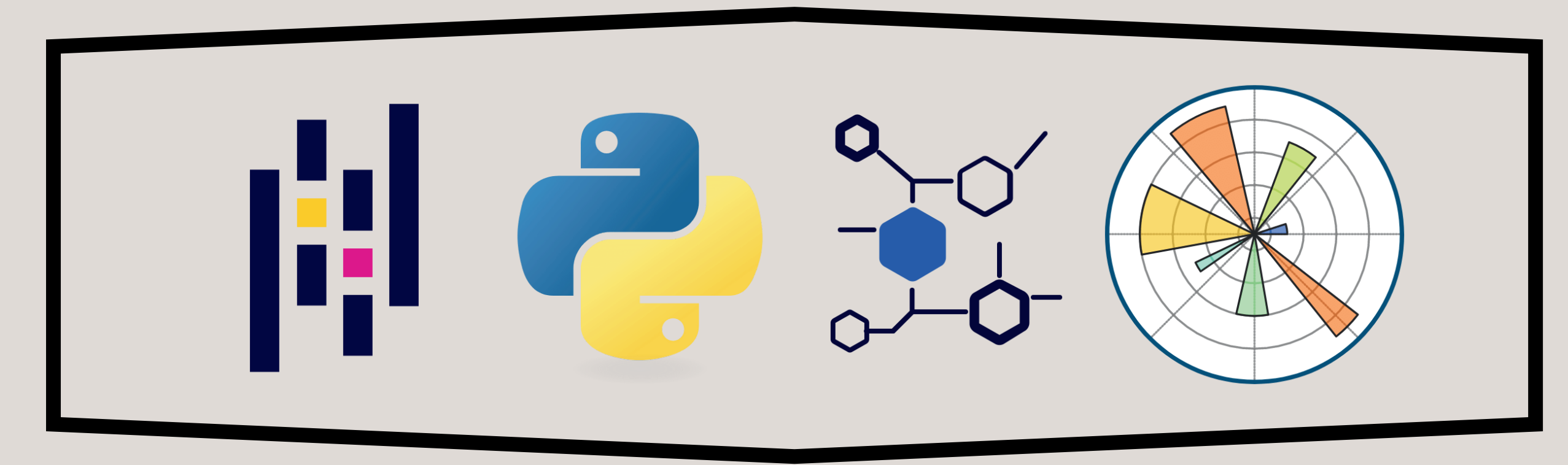
ClassSync AI automates the scheduling process using a Genetic Algorithm-based optimizer. It generates a conflict-free, resource-efficient master timetable that is visually clean, color-coded, and segmented by sections, instructors, and rooms.

CORE ALGORITHM

Uses a multi-generation Genetic Algorithm with adaptive mutation, elitism, and fitness scoring based on coverage, conflict avoidance, session balancing, and placement quality.

Saad Mughal
F2023-009

TOOLS & TECHNOLOGIES



SAMPLE OUTPUT

Generated by ClassSync AI

Day	Room	08:00 - 09:30	09:30 - 11:00	11:00 - 12:30	12:30 - 14:00	14:00 - 15:30
Tuesday	SB 004					Artificial Intelligence B CS4 Ms. Noreen Khalid
Thursday	SLASS 015		Artificial Intelligence C CS4 Ms. Noreen Khalid			
	SB 205			Artificial Intelligence B CS4 Ms. Noreen Khalid		
Friday	SVAD 403 Lab	Artificial Intelligence Lab C CS4 Ms. Noreen Khalid	Artificial Intelligence Lab C CS4 Ms. Noreen Khalid			
	SLASS 013					Artificial Intelligence C CS4 Ms. Noreen Khalid

KEY FEATURES

- Smart fallback for missed labs
- Instructor load balancing
- Section/Instructor/Room breakdowns
- Color-coded Excel outputs with watermark
- Configurable export settings

CALL TO ACTION

Experience the future of academic scheduling. Want conflict-free, optimized timetables for your institution? Let's connect and bring ClassSync AI to your campus.

Sheikh Muhammad Ibrahim
F2023-630